

Robot That Breathes Life Into AC Ducts

DuctBot can inspect and easily travel inside the sleek confines of the air-conditioner ducts and cleanse them of all the dust and grime, enhancing the energy efficiency of the air-conditioner



DuctBot kit

VANDANA SHARMA

"To regard old problems from a new angle requires creative imagination and marks real advance in science."

— Albert Einstein

For commercial set-ups such as shopping malls and office complexes, which mostly have centrally air-conditioned environments, getting air-conditioners' ducts cleaned has hitherto remained a challenge. The reasons are multiple. The equipment required to clean the ducts of these air-conditioners (ACs) are extremely expensive and bulky.

Manually doing this job is tedious and risky for the health of the person getting into these ducts. Considering that the ducts have a narrow passage, invariably, children are put into these dust-filled tunnels for cleaning—a practice that's illegal.

Apart from these issues, the choking of AC ducts over time affects the quality of the air circulated by the AC. It also leads to more power consumption and in some cases even breakdown of the air-conditioning system.

To address these problems, RoboSoft Systems has developed DuctBot—a toy car-sized robot that measures 23 cm in length, 19 cm in width and

9 cm in height. Priced at Rs 100,000, the device is claimed to cost five times less than the versions available globally.

Some vital statistics

DuctBot is a skid-steer drive gadget weighing 2 kg. It includes high-intensity LED light and charge-coupled device (CCD) colour camera to inspect the duct and provide a live feed. The vehicle moves on wheels and has attachments for cleaning the duct. It is fitted with a motor that can be controlled and switched on and off by supplying differential voltage. The device can be controlled wirelessly using Sony PS2 joystick.

A small slot in the duct is enough to put compact DuctBot inside and inspect air ducting condition of the flooring. DuctBot can go into spaces as tight as 9.6×17.8 cm² (4×7 square inches) in cross section. Once it is inside the air duct from one end, it blows compressed air in the duct helping the filter attached at the other end to collect the dust.

A four-member team worked on building the device, which took four years of R&D. "The device can go

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